When to use)

- finding the top k smallest/largest elements.
- finding the kth smallest/largest element.
- find the k most frequent element.

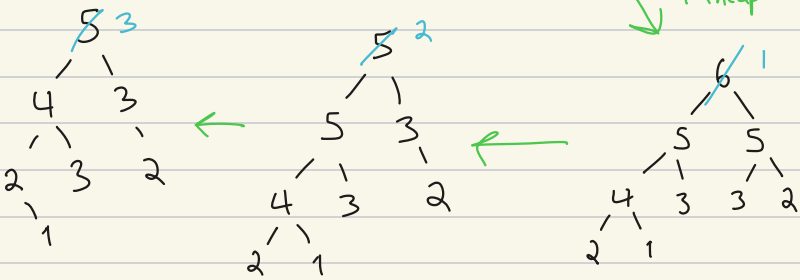
What is it)

- this method is using a heap to store the list we are given. We can then pop and push elements. So instead of sorting, we can use it to hold the smallest/largest K values, then for every element check to see if we should pop or push.

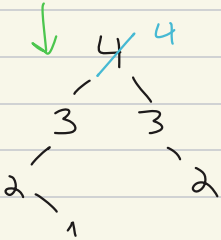
k=4

nums = [3, 2, 3, 1, 2, 4, 5, 5, 6]

transform into maxheap $O(n)$



pop k times $O(\log n)$



final pop == k
return it

Why use it)

- this improves the time complexity from $O(n \log n)$ to $O(n + k \log n)$ where $k < n$